

# CS & IT ENGINEERING



## COMPUTER ORGANIZATION AND ARCHITECTURE

### Cache Organization

Lecture No.- 6

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# Recap of Previous Lecture



Topic

Cache Initialization

Topic

Set Associative Mapping

Doubt ?





# Topics to be Covered



Topic

Set Associative Mapping

Topic

Fully Associative Mapping

Topic

Block Replacement

Topic

Miss Penalty

#Q. Consider a 64 bytes direct mapped cache with a block size of 16 bytes. Main memory size is 256 bytes. Currently in the cache, the blocks are having tags as follows:

|                        | Block | Tag |
|------------------------|-------|-----|
| $(1000)_2 = (8)_{10}$  | 00    | 10  |
| $(0101)_2 = (5)_{10}$  | 01    | 01  |
| $(1110)_2 = (14)_{10}$ | 10    | 11  |
| $(0111)_2 = (7)_{10}$  | 11    | 01  |

| Tag | block | byte |
|-----|-------|------|
|-----|-------|------|

$\xrightarrow{2} \xrightarrow{2} \xrightarrow{4}$   
 mm block no.

Identify the correct statement with respect to the availability of the main memory data into cache?

- a) ☒ Main memory byte number 243 present in cache
- b) ☒ Main memory byte number 143 present in cache
- c) ☒ Main memory byte number 43 present in cache
- d) ☒ Main memory byte number 119 present in cache

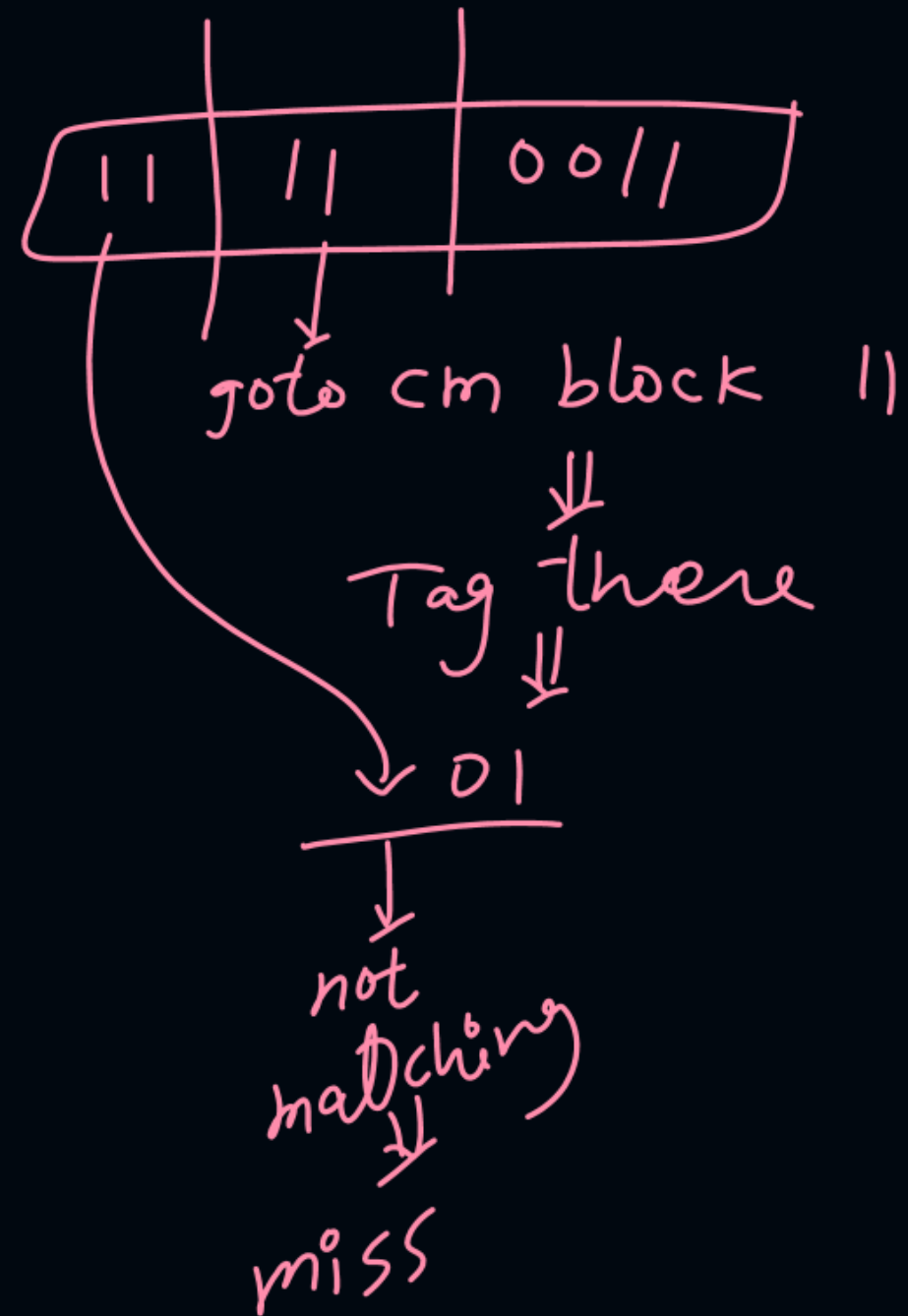
$$\text{mm block no.} = \frac{143}{16} = 8$$

$$= \frac{43}{16} = 2$$

$$\frac{119}{16} = 7$$



option (a)  $\Rightarrow (243)_{10} = (11110011)_2$



mm block no.  $= \left\lfloor \frac{243}{16} \right\rfloor$

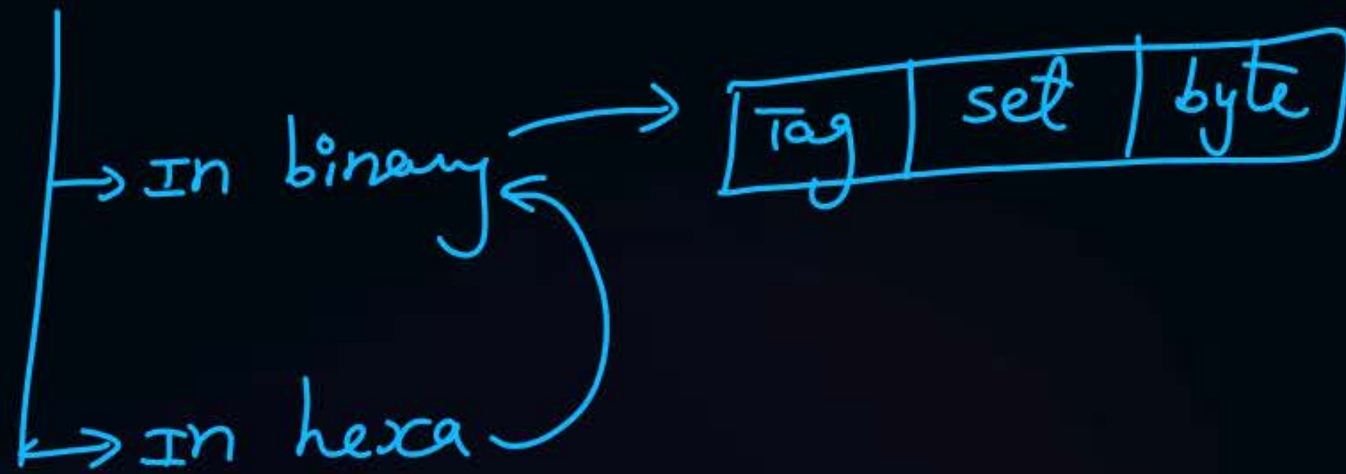
$= 15$

not present in cache



## Topic : Calculating CM Set Number from MM Address

mm add.



given in decimal:-

$$\text{mm block no.} = \left\lfloor \frac{\text{mm address}}{\text{block size}} \right\rfloor$$

$$\text{byte offset} = \text{mm add.} \% \text{ block size}$$

$$\text{cm set no.} = (\text{mm block no.}) \% \text{ no. of sets in cache}$$

$$\text{Tag} = \left\lfloor \frac{\text{mm block no.}}{\text{no. of sets in cache}} \right\rfloor$$

Ques)

mm add. = 16 bits

Cache size = 512 B

block size = 8 B

4-way set ass.

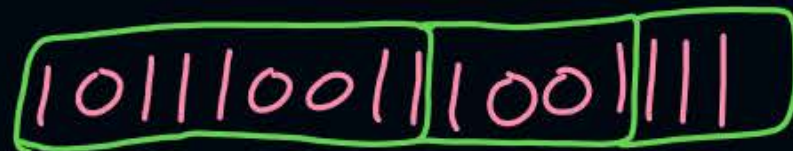
← 16 →



no. of sets =  $2^4 = 16$

$$\begin{aligned} &\leftarrow \log 512 - \log 4 \rightarrow \\ &= 9 - 2 \\ &= 7 \end{aligned}$$

mm add.



Tag      set      byte

↓

$(1001)_2$

$(9)_{10}$

mm address =  $(959)_{10}$

$$\text{mm block no.} = \left\lfloor \frac{959}{8} \right\rfloor = 119$$

byte offset = 7

$$\text{cm set no.} = 119 \% 16 = 7$$

$$\text{Tag} = \left\lfloor \frac{119}{16} \right\rfloor = 7$$



## [MCQ]



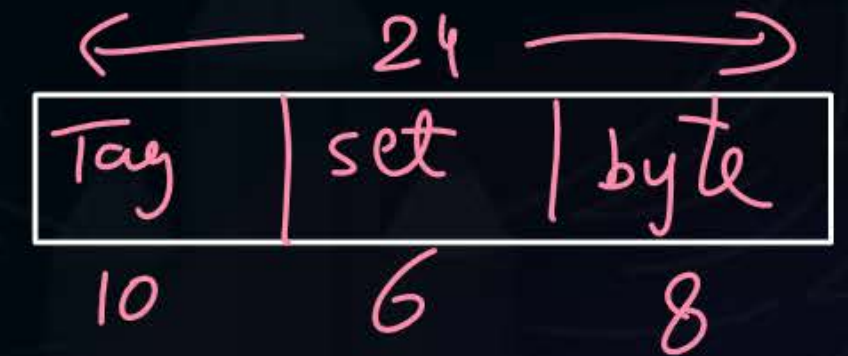
#Q. A computer system with a word length of 32 bits has a 16 MB byte- addressable main memory and a 64 KB, 4-way set associative cache memory with a block size of 256 bytes. Consider the following four physical addresses represented in hexadecimal notation.

A1 = 0x42C8A4 → 1100010000 set no.  
A2 = 0x546888 → 011010000 set no.  
A3 = 0x6A289C → 001010000 set no.  
A4 = 0x5E4880 → 010010000 set no.

Which one of the following is TRUE?

- ☒ A A1 and A4 are mapped to different cache sets.
- ☒ B A2 and A3 are mapped to the same cache set.
- ☒ C A3 and A4 are mapped to the same cache set.
- ☒ D A1 and A3 are mapped to the same cache set.

$$2^{24} \text{ B} \Rightarrow \text{mem add} = 24 \text{ bits}$$



$$\begin{aligned} &\leftarrow \log 64\text{K} - \log 4 \rightarrow \\ &\quad \downarrow \\ &\quad 16 - 2 \\ &\quad \leftarrow 14 \rightarrow \end{aligned}$$





# Topic : Fully Associative Mapping

Direct mapped

|   |      |
|---|------|
| 1 | 1825 |
|   |      |
|   |      |
|   |      |

2-way

|   |    |    |
|---|----|----|
|   |    |    |
| 1 | 18 | 25 |
|   |    |    |

4-way

|   |   |   |    |
|---|---|---|----|
|   |   |   |    |
| 1 | 3 | 5 | .. |
|   |   |   |    |

no. of blocks  
in cache - way



# Topic : Fully Associative Mapping

no. of blocks in cache = 64

$$\log_2 1 = 0$$

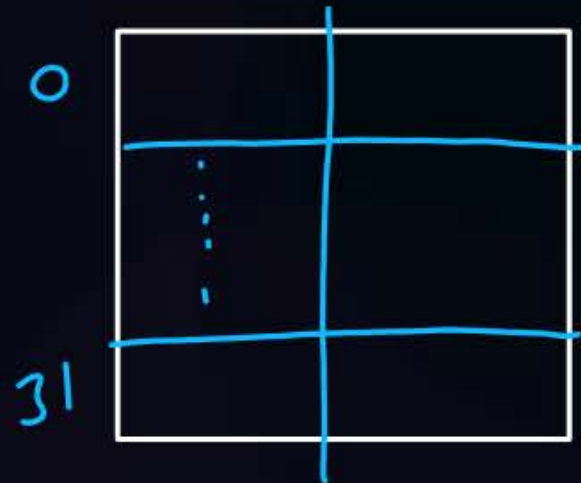
Direct mapped

| Tag | block | byte |
|-----|-------|------|
| y   | 6     | x    |



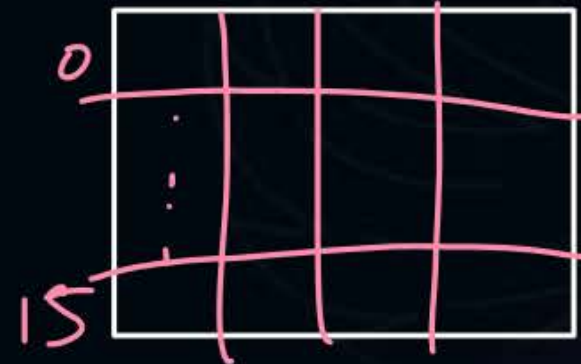
2-way

| Tag | set | byte |
|-----|-----|------|
| y+1 | 5   | x    |



4-way

| Tag | set | byte |
|-----|-----|------|
| y+2 | 4   | x    |



64-way

| Tag | set | byte |
|-----|-----|------|
| y+6 | 0   | x    |



single set in cache



mem add.



→ mem block no. is Tag

Index ⇒ 0 bit

Tag directory size = no. of blocks in cache \* (Tag + extra bits)

## [Question]

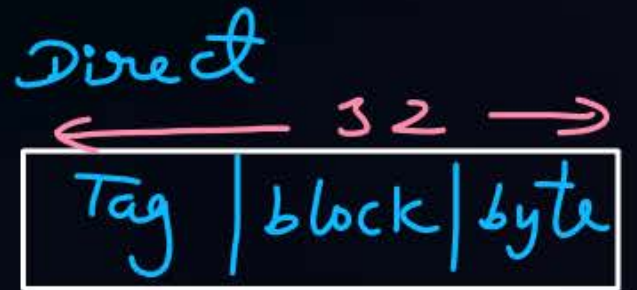


#Q. Cache size = 256KB

MM address = 32 bits

Block size = 32 Bytes

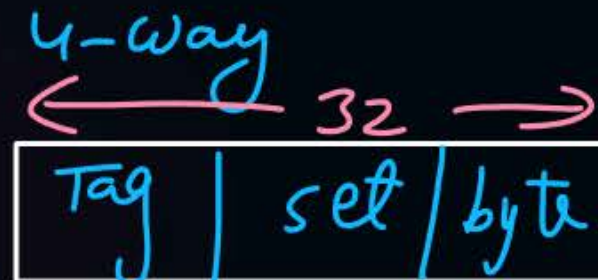
Tag and Tag directory for direct, 4-way set associative and fully associative



14      13      5

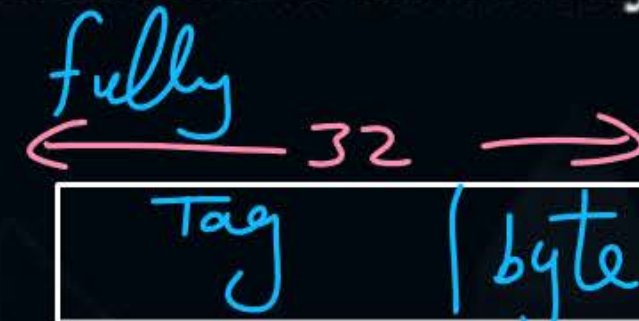
$\log 256 = 18$

Tag directory size  
 $= 2^{13} * 14 \text{ bits}$



16      11      5

$= 2^{13} * 16 \text{ bits}$



27      5

$= 2^{13} * 27 \text{ bits}$



|       | max    | min    |
|-------|--------|--------|
| Tag   | fully  | Direct |
| Index | Direct | fully  |

ques fully ass. Cache has no. of blocks =  $2^{12}$   
mm has no. of block =  $2^{20}$   
Cache Tag directory size = — ?

sol<sup>n</sup> mm blockno.  $\Rightarrow$  Tag = 20 bits

$$\text{Tag directory size} = \underline{\underline{2^{12} * 20 \text{ bits}}} \quad \text{Ans.}$$

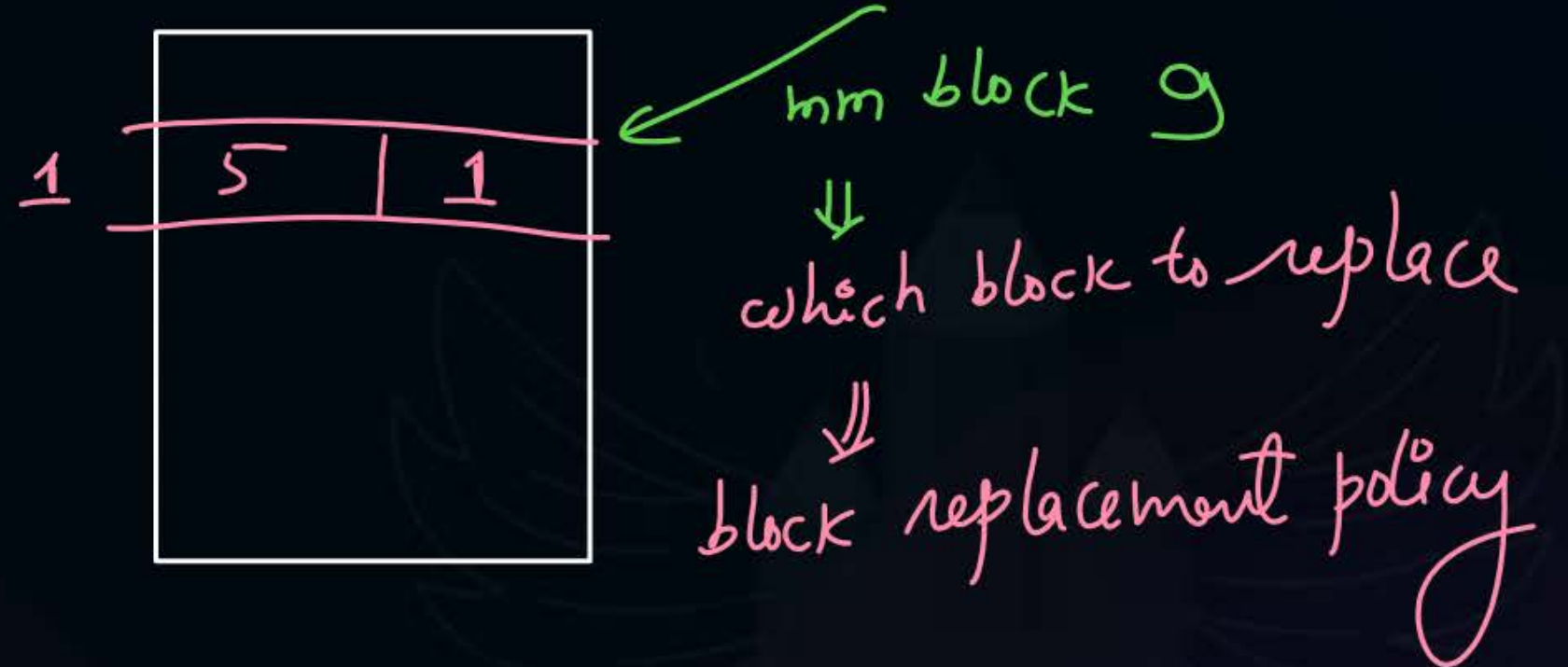
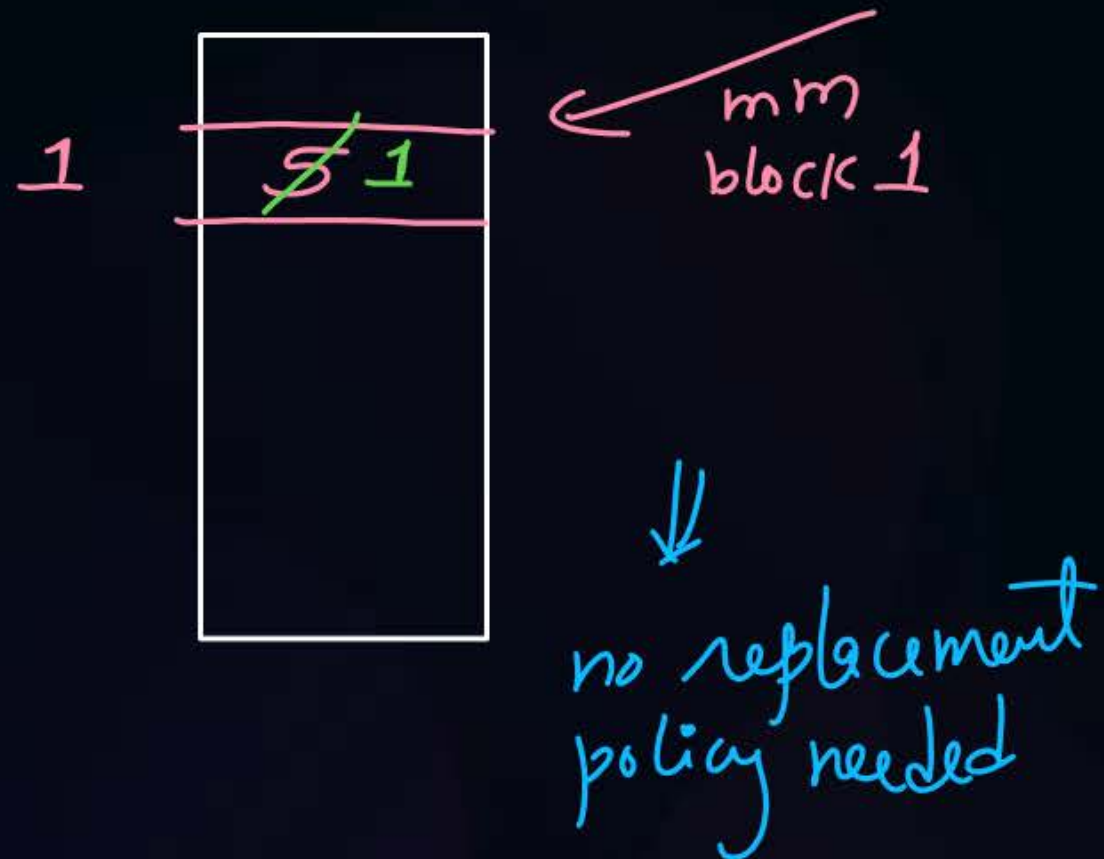




## Topic : Block Replacement

Direct mapping

2-way set ass.



## Block replacement policies

1. FIFO

2. optimal

3. LRU (Least Recently used)

→ The block which has not be used since longest period of time in past.

#Q. Consider a small two-way set-associative cache memory, consisting of four blocks. For choosing the block to be replaced, use the least recently used (LRU) scheme. The number of cache misses for the following sequence of block addresses is:

8, 12, 0, 12, 8  
 ✓ ✓ ✓ *hit* ✓

**A** 2

**B** 3

**C** 4

**D** 5

$$\text{no. of sets} = \frac{4}{2} = 2$$



$$8 \% 2 = 0$$

$$12 \% 2 = 0$$

$$0 \% 2 = 0$$



#Q. Consider a Direct Mapped Cache with 8 cache blocks (numbered 0-7). If the memory block requests are in the following order

✓ 3, ✓ 5, ✓ 2, ✓ 8, ✓ 0, ✓ 63, ✓ 9, ✓ 16, ✓ 20, ✓ 17, ✓ 25, ✓ 18, ✓ 30, ✓ 24, ✓ 2, ✗ 63, ✗ 5, ✓ 82, ✓ 17, ✗ 24.

Which of the following memory blocks will not be in the cache at the end of the sequence?

**A**

3

**B**

✓ 18

**C**

20

**D**

30

cache

|   |                                                        |
|---|--------------------------------------------------------|
| 0 | <del>8</del> <del>0</del> <del>16</del> <del>24</del>  |
| 1 | <del>9</del> <del>17</del> <del>25</del> <del>17</del> |
| 2 | <del>2</del> <del>18</del> <del>2</del> <del>82</del>  |
| 3 | 3                                                      |
| 4 | 20                                                     |
| 5 | 5                                                      |
| 6 | 30                                                     |
| 7 | 63                                                     |

mem block no.

/ no. of blocks in cache

$$3 \% 8 = 3$$

no. of hits = 3  $\Rightarrow$  hit ratio =  $\frac{3}{20}$   
 no. of miss = 17

$$\text{miss ratio} = \frac{17}{20}$$

#Q. Consider a 4-way set associative cache (initially empty) with total 16 cache blocks. The main memory consists of 256 block and the request for memory blocks is in the following order:

0, 255, 1, 4, 3, 8, 133, 159, 216, 129, 63, 8, 48, 32, 73, 92, 155

Which one of the following memory block will not be in cache if LRU replacement policy is used?

**A**

3

**B**

8

**C**

129

**D**

216



#Q. Consider a fully associative cache with 8 cache blocks (numbered 0–7) and the following sequence of memory block requests:

4,3,25,8,19,6,25,8,16,35,45,22,8,3,16,25,7

If LRU replacement policy is used, which cache block will have memory block 7?

**A**

4

**B**

5

**C**

6

**D**

7





## 2 mins Summary



**Topic**

Set Associative Mapping

**Topic**

Fully Associative Mapping

**Topic**

Block Replacement

**Topic**

Miss Penalty



**Happy Learning**

**THANK - YOU**

